C4 Chemical Changes – Knowledge Organiser

AQA GCSE Chemistry • Set 1: Reactivity and Extraction Paper 1

#	Question	Answer
1	Reactivity series (most reactive first)	K, Na, Li, Ca, Mg, C, Zn, Fe, H, Cu
2	Products of a metal with water	Metal hydroxide + hydrogen
3	Products of a metal with acid	Salt + hydrogen
4	Oxidation (in terms of oxygen)	Gain of oxygen
5	Reduction (in terms of oxygen)	Loss of oxygen
6	Metals extracted by reduction with carbon	Those less reactive than carbon, e.g. iron
7	Metals extracted by electrolysis	Those more reactive than carbon, e.g. aluminium
8	Metal found uncombined in the Earth	Gold
9	Displacement reaction	More reactive metal displaces a less reactive one from a compound
10	Ore	Rock containing enough metal to extract

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AQA GCSE Chemistry • Set 2: Redox and Salts **Paper 1**

#	Question	Answer
1	Oxidation (in terms of electrons)	Loss of electrons
2	Reduction (in terms of electrons)	Gain of electrons
3	Spectator ion	Ion that does not change in the reaction
4	Acid + alkali	Salt + water
5	Acid + metal oxide	Salt + water
6	Acid + metal carbonate	Salt + water + carbon dioxide
7	Salts made from hydrochloric acid	Chlorides
8	Salts made from sulfuric acid	Sulfates
9	Salts made from nitric acid	Nitrates
10	Base	Substance that neutralises an acid

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AQA GCSE Chemistry • Set 3: pH and Strong and Weak Acids **Paper 1**

#	Question	Answer
1	pH of a neutral solution	7
2	pH of acids	Below 7
3	pH of alkalis	Above 7
4	Ion that makes solutions acidic	H^+
5	Ion that makes solutions alkaline	OH^-
6	Ionic equation for neutralisation	$H^+ + OH^- \rightarrow H_2O$
7	Strong acid	Completely ionised in water
8	Weak acid	Partially ionised in water
9	Examples of strong acids	Hydrochloric, nitric, sulfuric
10	Change in $[H^+]$ when pH falls by 1	× 10

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AQA GCSE Chemistry • Set 4: Electrolysis **Paper 1**

#	Question	Answer
1	Electrolysis	Splitting a compound using electricity
2	Electrolyte	Molten or dissolved ionic compound
3	Cathode	Negative electrode
4	Anode	Positive electrode
5	Ions attracted to the cathode	Positive ions
6	Products of electrolysis of molten lead bromide	Lead and bromine
7	Product at the cathode in aqueous solution	Hydrogen, unless the metal is less reactive than hydrogen
8	Product at the anode in aqueous solution	Oxygen, unless halide ions are present
9	Products of electrolysis of brine	Hydrogen, chlorine, sodium hydroxide
10	Half-equation for chloride ions at the anode	$2Cl^- \rightarrow Cl_2 + 2e^-$

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AQA GCSE Chemistry • Set 5: Electrolysis of Aluminium Oxide **Paper 1**

#	Question	Answer
1	Aluminium is extracted from	Aluminium oxide
2	What aluminium oxide is dissolved in	Molten cryolite
3	Why cryolite is used	Lowers the melting point, saving energy
4	Material of the electrodes	Carbon (graphite)
5	Product at the cathode	Aluminium
6	Product at the anode	Oxygen
7	Why the anodes must be replaced	Carbon reacts with oxygen to form CO_2
8	Half-equation at the cathode	$Al^{3+} + 3e^- \rightarrow Al$
9	Half-equation for hydrogen at the cathode	$2H^+ + 2e^- \rightarrow H_2$
10	Half-equation for oxygen at the anode	$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$

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AQA GCSE Chemistry • Set 6: Titrations Paper 1 Triple only

#	Question	Answer
1	Purpose of a titration	Find volumes of acid and alkali that exactly react
2	Indicators for a titration	Phenolphthalein or methyl orange
3	Why universal indicator is not used	Gradual colour change
4	Phenolphthalein colours: acid / alkali	Colourless / pink
5	Methyl orange colours: acid / alkali	Red / yellow
6	Why a white tile is used	To see the colour change clearly
7	How acid is added near the end point	Dropwise
8	Where the burette is read	Bottom of the meniscus
9	Concordant results	Within 0.10 cm^3 of each other
10	Burette is rinsed with	The acid it will contain